CE4

Project Introduction

CE4.1

Project Name	MUSC2273 Electronic Music: Generative Process Midterm Project: Theremin
Project Duration	01/2022 to 06/2022
Project Organization	The University of Western Australia
My Title	Electronics Engineer

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What is Theremin: The Theremin, a famous electronic musical instrument invented in 1920, is played without any physical contact. The performer controls the sound by moving their hands in the air around two antennas. This non-contact interaction allows for expressive glissandi (sliding between notes) and vibrato, contributing to the instrument's distinctive, often ethereal or sci-fi sound heard in various musical pieces and film scores.

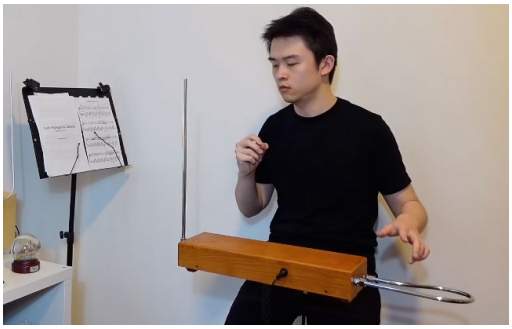


Figure 1: Commercial Theremin operation: left antenna controls the tone, right controls the volume.

This project details the design and construction of a functional Theremin. My work focused on implementing the core circuitry necessary to sense hand movements near antennas and translate these movements into controllable audio signals, finally played out with a speaker.

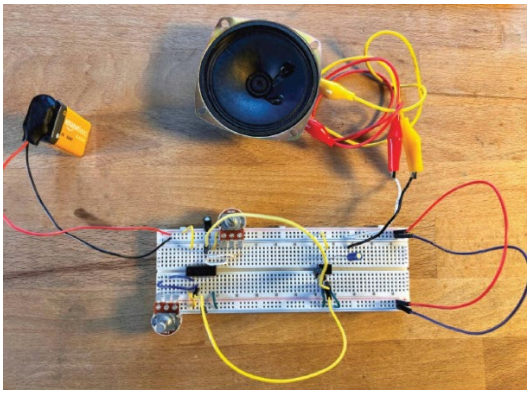


Figure 2: The Theremin I constructed, with basic functionality demonstrated. The design of physical enclosure is excluded (outside MUSC2273 scope).

Project Background

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This is an individual project of the MUSC2273 (mid-term project). Students were asked to submit an art design related to the music and electronics. Consequently, both the design concept (focused on 8-bit sound quality Theremin) and the subsequent circuit were entirely my own work.

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The main objective of this project is to design an 8-bit audio format Theremin.

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- For the software simulation phase, it was necessary to understand the principles of 8-bit encoding and electrical signals. This involved selecting and assembling components from the limited library available within the simulation software to build the circuit virtually.
- The physical implementation phase required selecting suitable integrated circuits components (e.g., digital chips) and familiar with practical skills, such as soldering, choosing the correct components for breadboard assembly, and physically constructing the circuit.

Project Activities

CE4.6 Overview

- The first phase involved **software simulation** to design and verify the core electronic principles of the Theremin. This included modeling circuits for sensing hand capacitance and subsequent pitch generation (VCOs) and volume control. The simulation confirmed how capacitance variations translated into changes in output frequency (i.e., the pitch, range from 200 Hz to 2 kHz) and amplitude (i.e., the volume). While limited in modeling real-world Theremin, this phase was crucial for validating the control logic and signal flow prior to physical construction.
- Following successful simulation, **physical circuit implementation** took place on a breadboard using selected ICs for capacitance sensing, signal processing, and sound generation. The assembled circuit was powered by a 9V stable DC supply, with the circuit output connected to an audio amplifier to drive a small speaker. Implementation required practical skills including component selection, careful breadboard layout, and basic soldering for the DAC. Debugging involved verifying power/signals, troubleshooting connections, and calibrating the circuit's sensitivity for playable pitch and volume control.

CE4.7 Encoding 8-bit Signal

I designed a 4-bit binary encoding logic to control audio output, enabling accurate representation of seven musical notes within a single octave. To ensure a clear control mechanism, I set the first bit as a flag to indicate whether a note is active, using "1xxx" for active and "0xxx" for inactive, and specifically assigned "0000" to turn off the speaker for silence. To allow the system to accurately play or stop sound based on input signals, I mapped each note status to a corresponding binary

code and aligned these with the sequence of natural musical notes, thereby establishing a logical structure consistent with the natural keys of a piano, hence ensuring both stability in note switching and accuracy in circuit decoding and control.

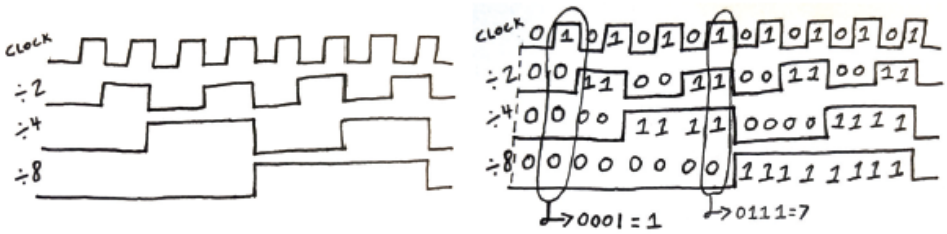


Figure 3: How to generate 4-bit audio signal with one chip.

I used two digital chips to construct the core of the Theremin. The first chip was responsible for generating a 4-bit control signal to select different musical notes based on the previously designed encoding logic. To implement volume control, I added a second chip specifically for adjusting the loudness. Since the note selection and volume control were handled separately, I decided to combine their output signals into an 8-bit signal to meet the overall audio output requirements of the system. Considering that 8-bit audio was widely used in 20th-century audio systems and enabled sound effects in games like Super Mario and Pokémon on the Gameboy, I adopted this approach to ensure the audio quality met the acceptable level for the project.

CE4.8 Software Simulation

Speaker & Amplifier: During circuit simulation using TinkerCAD (a free-to-use web app), I encountered limitations simulating the antenna and associated encoding circuitry, due to the restrictions of the free version. To address this challenge, I employed two standard 555 timer ICs to simulate the outputs expected from the antenna reception and encoding stages. I then prioritized building and successfully testing the audio amplifier and speaker section using the virtual breadboard and necessary components. Subsequently, the actual antenna and encoding circuitry will be integrated to replace these 555 chips for signal generation, providing the encoding for the tone and volume.

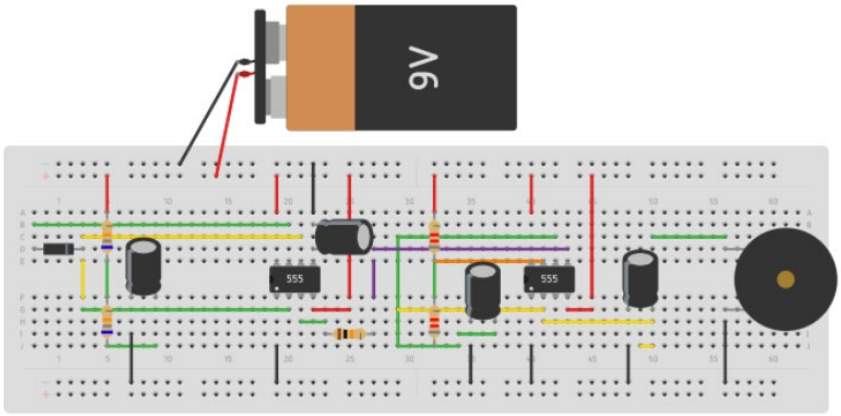


Figure 4: TinkerCAD virtual breadboard to test the functionality of the amplifier and speaker.

Sound Provider (ICs & Antenna): To simulate the octave harmonizer, which providing the tone of the Theremin, I used Fritzing software to create the schematic. The main electronic components included the CD4093 and CD4040 ICs, resistors, 100nF capacitors, and 100kΩ potentiometer.

- To build the core sound-generating module of the Theremin, I decided to use the **CD4040** as the main frequency divider chip, utilizing its internal cascaded binary counter to effectively divide the input high-frequency clock signal into frequencies audible to the human ear. After confirming that the chip provided multiple output pins, each halving the frequency of the previous stage, I evaluated the frequency ranges of these outputs and ultimately selected eight outputs from Q0 to Q7 to represent eight different pitch signals, thereby constructing a basic scale system. I first tested the compatibility between the CD4040 and an external oscillator in a simulation environment to ensure the high-frequency clock signal could be accurately recognized and stably divided by the chip. Then, I designed the circuit connections, linking the outputs of the CD4040 to the speaker module so that different pitches would be produced depending on the selected output pin, and incorporated a 4-bit encoding system to dynamically switch between pitch signals. During the physical circuit implementation, I paid special attention to aligning the divided frequency outputs with the system's overall 8-bit audio structure. Considering the audio signal required 256 levels of variation (2^8), I selected the Q7 pin of the CD4040 as the main reference frequency and coordinated the pitch output module design accordingly. While wiring, I built the prototype circuit on a breadboard, using jumper wires to connect each divided output pin to the control unit and speaker, ensuring stable connections for power, ground, and signal input. To validate the design, I used an oscilloscope to check the waveform signals at Q7 and other output pins, confirming that their frequencies matched the expected division ratios, thereby ensuring the accuracy and consistency of the audio output.
- I selected the **CD4093** as a square wave oscillator to generate the clock signal that drives the CD4040 frequency divider, thereby enabling fundamental control over the pitch range of the Theremin. To set an appropriate oscillation frequency, I adjusted the values of the resistor and capacitor based on circuit requirements, ensuring that the output frequency met the desired pitch range. Since this clock signal directly affects the output frequency of each divided channel in the CD4040, I focused on how the oscillator's frequency influences the subsequent pitch generation, and accordingly optimized the circuit parameters to ensure the overall pitch control system operated stably and responsively.

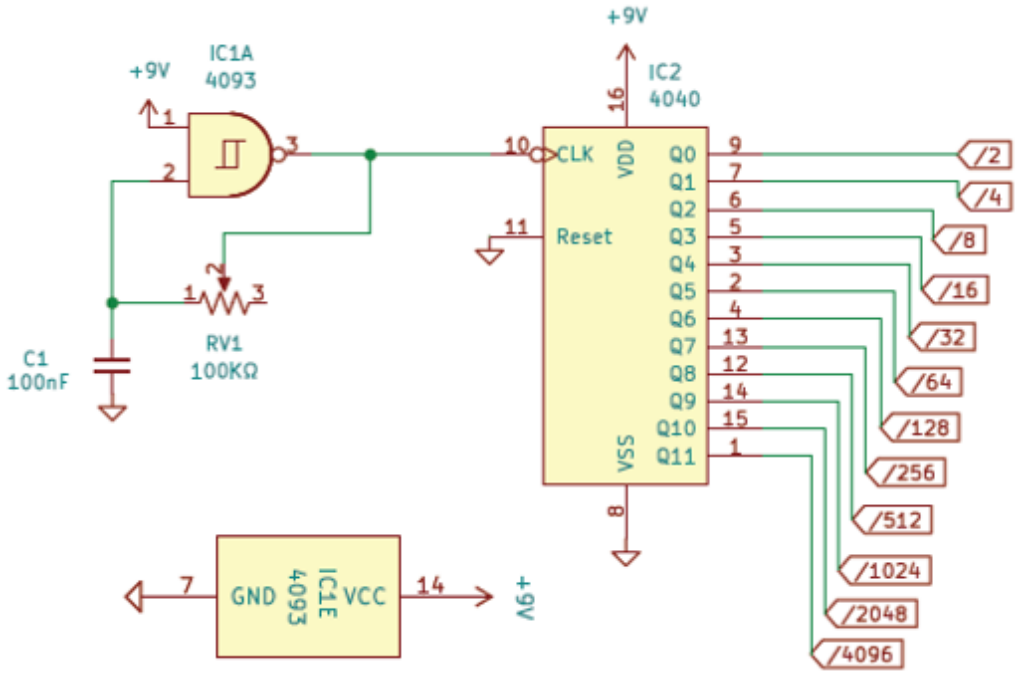


Figure 5: Circuit of our Theremin's octave harmonizer, created via the Windows App - Fritzing

- The 100kΩ potentiometer is used to simulate the **Theremin's antenna**, and in practice, it's replaced by a **light-dependent resistor** to let hand movement control the pitch. Together with the 100nF capacitor, it forms a high pass filter that blocks DC while allowing AC signals to pass, enabling the harmonizer to function properly. This filter also smooths the output by attenuating overly high-frequency signals, keeping the theremin's tone within a pleasant range (0-4 kHz).

CE4.9 Hardware Implementation

After completing the software simulation and verifying the correctness of the circuit logic, I proceeded to transform the design into a physical circuit. Based on the logical structure and component interactions confirmed during simulation, I selected suitable integrated circuit (IC) chips and other necessary electronic components. Then, I assembled the circuit on a breadboard, carefully translating the virtual connections from the simulation into actual wire connections to ensure the physical setup matched the simulated design precisely. Notably, the speaker shown in Figure 6 was the only component in the entire breadboard design that required soldering. Its functionality was verified using a computer audio cable to ensure it was working correctly before integration. The complete assembled circuit on the breadboard is presented in Figure 7.



Figure 6: Speaker *Pre-Diagnosis and Soldering*

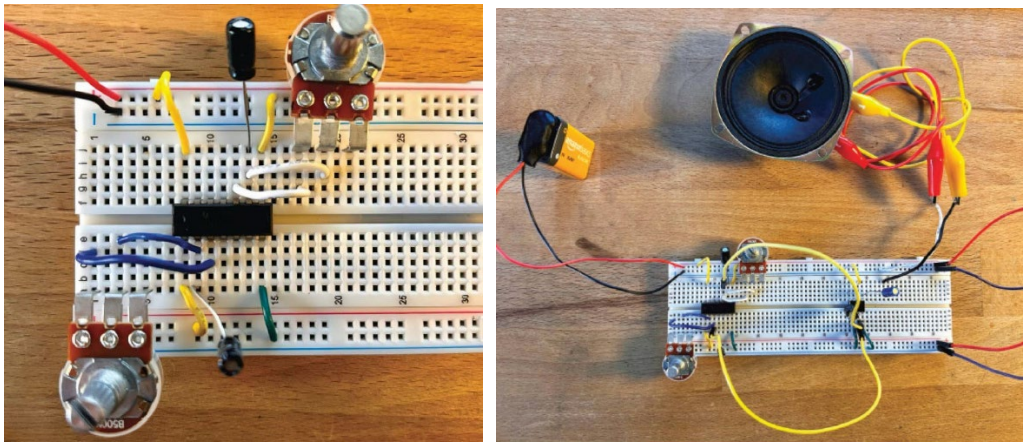


Figure 6: Physical implementation of my Theremin on Breadboard.

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Since the focus of this project was on functional circuit implementation, I did not design a physical enclosure for the Theremin, as the acoustic modeling of instrument housings was beyond the scope of the MUSC2273 course requirements and outside my technical familiarity at the time. Likewise, in this initial prototype phase, I decided not to use traditional Theremin antennas. Instead, I used light-dependent resistors as simulated sensor inputs to demonstrate the core principles of pitch and volume control. Based on the course objectives and the feasibility at this development stage, I made these simplification choices to ensure successful demonstration of the essential functions.

Project Summary

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I paid strict attention to the project phases, ensuring the processes of both simulation and physical implementation were conducted diligently. While debugging the circuit on the breadboard, I made sure that I carefully checked connections between the various components and the operation of each component itself. I carried out thorough research on the relevant technical principles to ensure that my design was in line with the fundamental concepts of Theremin operation and 8-bit audio generation. Besides, special attention was paid to the practical challenges of physical implementation on a breadboard. This process involved careful component selection and layout, which not only tested the design's robustness in a real-world environment but also developed essential troubleshooting skills during assembly.